Teacher Print Master

Grade 5 OpenSciEd Unit 5.1 | Lesson 1

ONE teacher copy | Single-sided | 52 pages including this cover.
Source: 5.1 Teacher Guide, Version 1.0, Lesson 1 printed pages 50-100
(source PDF pages 56-106). This is a source-page extract; publisher pages are unchanged.
The master includes lesson sequence, setup, learning goal, pre-assessment guidance,
lesson facilitation, and Lesson 1 teacher references. Keep teacher-only.
Official 5.1 Lesson 1 slide deck is not included; obtain it before teaching the image/video steps.
Print the cover and following pages single-sided. Original source page labels are retained.

Cover/print map. Original publisher pages follow; retain their attribution and labels.

LESSON 1

How does a fallen tree become a nurse log?

What we do:

In this Anchoring Phenomenon Lesson, we observe pictures and video of fallen trees that are supporting many other things in the forest to live and grow. We learn that scientists call these nurse logs. We model to explain our initial ideas about how a fallen tree becomes a nurse log and create an initial consensus model. We brainstorm related phenomena. We share questions about nurse logs to create a Driving Question Board and ideas for investigations to pursue next.



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In the next lesson, we will observe a nurse log and identify the things we notice as components of the nurse log system. We will focus on moss as an important component. We will ask questions about and closely observe moss.

What we figure out:

- Nurse logs are fallen trees that help other things live and grow.
- We see lots of things on the nurse logs, and have some ideas about other things that are likely found in, under, and around nurse logs.

Building Toward Standards

This unit builds toward these NGSS PEs: 5-PS1-1, 5-PS3-1, 5-LS1-1, 5-LS2-1. See this matrix for the NGSS elements and CCSS ELA and Math connections.

Learning Plan Snapshot

	duration	slides	materials
EXPLORE			
1. **Make observations.** We observe images and video of a puzzling phenomenon and then share our individual observations and questions as a class.	30 minutes	A-L	Explore a New Phenomenon, chart paper, chart paper marker
SYNTHESIZE			
2. **Develop a model.** We model our initial ideas individually about how a nurse log helps other things live and grow, and participate in a Consensus Discussion to develop an initial consensus model.	30 minutes	M-R	Initial Model, chart paper, chart paper markers
CONNECT			
3. **Identify related phenomena.** In small groups and then as a class, we share our ideas about phenomena that we think relate to the nurse log.	20 minutes	S-T	Related Phenomena, large (at least 11x17 inches) piece of paper, markers, chart paper, chart paper marker
NAVIGATE			
4. **Ask questions.** As a class, we create a Driving Question Board of all of the questions we have about nurse logs and brainstorm how we could investigate some of those questions.	30 minutes	U-X	Related Phenomena, sticky note, marker, Investigation Ideas, Notice and Wonder chart, Initial Consensus Model, Related Phenomena chart, chart paper, chart paper marker

Total time for this lesson: 110 minutes

Lesson Materials and Preparation

Per student:	**Per small group:**	**Per class:**
• Explore a New Phenomenon • Initial Model • Related Phenomena • sticky note • marker	• large (at least 11x17 inches) piece of paper • markers • Investigation Ideas	• chart paper • chart paper marker • chart paper markers • Notice and Wonder chart • Initial Consensus Model • Related Phenomena chart

Preparation Checklist • Estimated time: 30 minutes

- ☐ Be ready to use 5.1 Lesson 1 Slides.
  - ☐ Review the images and video in the slides to orient to the media that students will be engaging with to observe nurse logs.
  - ☐ As noted in the Explore, you may want to print the images on **slide G** for students to get a closer look as they notice and wonder.
- ☐ **Follow the guidance for receiving and immediately unpacking the live organisms** as detailed in Unit Advanced Preparation.
- ☐ **Forming a Scientists Circle:** You will form a Scientists Circle in this and many future lessons. If your students do not already meet in a circle in your classroom, take time during this lesson to plan and practice logistics for forming, equitably participating in, and exiting that space so students can do it quickly and smoothly in the future. As students get more comfortable with a Scientists Circle, they can move between it and other activity structures more quickly. Ideally they need to be able to see both the slides and the class whiteboard, but if that is not possible, the whiteboard (or chart paper) will be more critical for the discussion than the slides. If you have back-to-back science classes, ask students to leave the chairs in the circle.You can find more information about Scientists Circles in the Teacher Handbook.
- ☐ Throughout the unit, students will use a science notebook and printed handouts to support their science learning. You will distribute the handouts as needed in each lesson (not all at once for the unit). You can find more information in the Teacher Handbook about how science notebooks support students' science sensemaking in this program. Options for setting up science notebooks include
  - ☐ having students use a sewn or spiral-bound notebook where they can tape handouts inside and also write directly onto the lined pages;
  - ☐ having students use a 3-ring binder where they can add 3-hole-punched handouts (and lined loose-leaf paper if you want them to write on that).
- ☐ Locate or print the Word Wall card(s) for this lesson (see the Lesson Vocabulary table). Discuss and post the words during the specified components of the lesson. Alternatively, plan to develop your own Word Wall cards in collaboration with your students. A suggested definition is provided in the teacher guide, but your class may use slightly different language.

- [ ] Copy lesson handouts as listed in the table above. Optional: trim or 3-hole punch the handouts for faster addition to students' science notebooks. Alternatively, the following handouts are simple enough that students can write them directly into their science notebooks:
    - [ ] Explore a New Phenomenon
    - [ ] Related Phenomena
    - [ ] Investigation Ideas
- [ ] Prepare the following charts on chart paper or in a digital chart space:
    - [ ] Notice & Wonder chart
    - [ ] Initial Consensus Model
    - [ ] Driving Question Board
    - [ ] Related Phenomena chart
    - [ ] Ideas for Investigation (optional)
- [ ] See the Example Unit Models and Charts teacher reference to understand the purposes of the charts your class will develop in this unit and plan for how and when you will display them.
- [ ] Send home the Welcome Letter anytime after you complete Lesson 1.

Lesson 1 Vocabulary

	EXPLORE	SYNTHESIZE	CONNECT	NAVIGATE
Unit-Specific Science Ideas These are words that students will use for sensemaking in this unit. We will continue using them throughout the unit, so post them on the Word Wall.	nurse log			
Science Practices and Concepts That Build over Time These are words best learned as students engage with them over time. We will revisit them across units and grade levels.	phenomenon	modeling		

Three-dimensional Learning Goal(s)

Using the lens of:	Students will:	To make sense of:	Lesson learning goal:
Scale, Proportion, and Quantity	Develop and use models	How fallen trees become nurse logs	**1.** Develop models to describe our initial ideas about how a very large fallen tree becomes a nurse log over time and provides matter and energy to other organisms.

Lesson Assessment Guidance

What will students do?	Where can I check for understanding?	How can I use this assessment information?
1. Develop models to describe our initial ideas about how a very large fallen tree becomes a nurse log over time and provides matter and energy to other organisms.	**Assessment type:** Pre-assessment **Where to check for understanding:** During the Consensus Discussion in the Synthesis and on the Initial Model handout. **What to look and listen for:** • Words, drawings, spoken descriptions, movement and/or gestures showing ◦ the nurse log giving matter and/or energy to other living things ◦ arrows or gestures to show relationships between the nurse log and other things ◦ relative spatial scale between the nurse log and other things in the environment ◦ relative timescale for a fallen tree to become a nurse log • See additional ideas to look and listen for in the discussion prompts provided in the lesson. ◦ Accept all student ideas. Students may identify some of the things in, on, and around nurse logs, but expect a range of ideas and names.	This is a pre-assessment opportunity. Therefore, do not take a grade or score. Use this opportunity to determine the support your students may need around the practice of modeling, and around ideas of scale, matter, and energy, which they will continue to develop throughout this unit.

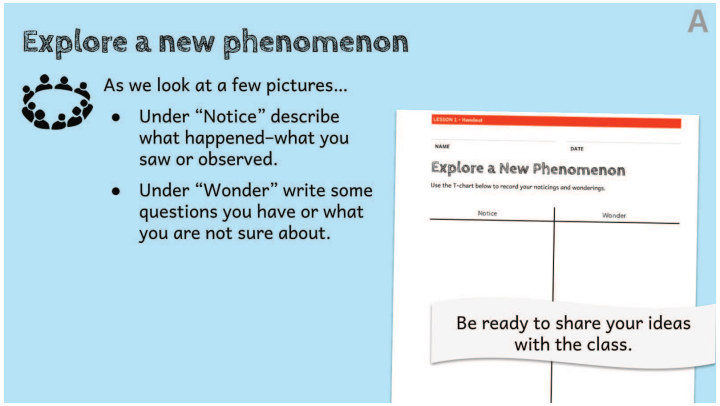
EXPLORE

Purpose: Introduce students to the unit's puzzling phenomenon of nurse logs and elicit all students' initial observations and questions to encourage students' engagement and participation in science, particularly for students who are underrepresented and undervalued in science.

1. Make observations. [30 min]

Introduce the unit phenomenon. Display **slide A**. Say something like, *I recently saw some pictures that made me say, 'Whoa. Cool.' I want to share them with you so we can explore the phenomenon shown in these pictures.*

- Define a **phenomenon** as *something that happens in the world*.



Observe a variety of pictures. Distribute the Explore a New Phenomenon handout to each student. Give students 1-2 minutes to individually jot their ideas on the handout as you move through **slides B-F** (note that **slide G** has all of the images on one slide if you prefer to show that). Emphasize that students can use a combination of words (in any language) and sketches.

Scale, Proportion, and Quantity

For the image on **slide E**, note with students how big the tree is, using the people next to the fallen and standing trees to get a sense of spatial scale. For the image on **slide F**, note with students how the picture shows a different perspective of a fallen tree: it is tipped over and we are seeing a side view of the bottom of the tree!

Broadening Access

It may be helpful to support students' processes of noticing and wondering by printing out **slide G** that has all of the images. Including visual representations of the phenomena as multiple means of representation will benefit students with diverse visual needs, as well as all students as they engage in a turn and talk with a partner. This provides multiple means of representation, especially to support students who are visually impaired.

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B

Make observations

Look at the picture.
What do you notice?
What do you wonder?



Wing-Chi Poon: CCBY 2.5

C

Make observations

Look at the picture.
What do you notice?
What do you wonder?



D

Make observations

Look at the picture.
What do you notice?
What do you wonder?



E

Make observations

Look at the picture.
What do you notice?
What do you wonder?



F

Make observations

Look at the picture.
What do you notice?
What do you wonder?



G

Make observations







You may want to have students turn and talk with a partner for each picture, or share some of their noticings and wonderings as a class. If you have students share, the goal is to surface a wide variety of student-generated ideas. You can encourage that variety by asking questions like

- Did anyone else notice that same thing?
- Did anyone notice something related to that?
- Did anyone notice something different?

You may also prompt students to share out something they heard a partner say that was interesting.

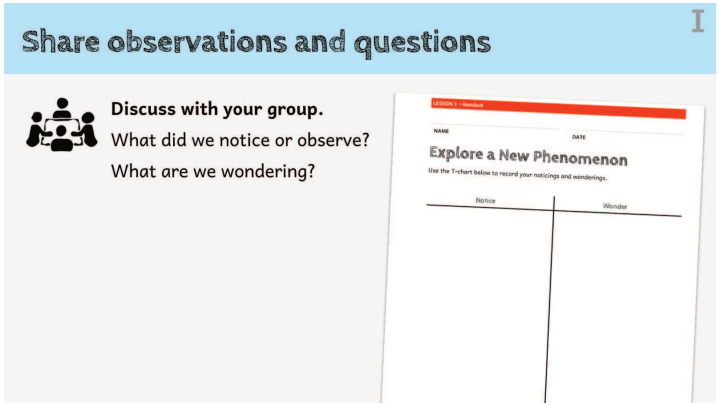
Show a video of a fallen tree. Display **slide H**. Introduce this video by saying something like, *I know from walking in the woods that when trees first fall, they don't look like the logs we saw on the previous slides. A friend sent me this video of a redwood tree that had recently fallen so we could take a closer look.* Play the video at least once, and likely a second time to help students home in on observations. As with the pictures above, emphasize the spatial scale in the video and the large size of the fallen tree.



Video of a fallen redwood tree

After the video, prompt students to add any additional noticings or wonderings to their charts.

Share observations and questions. Display **slide I.** Let students know they have 2-3 minutes to share their ideas with their table groups. They could use gestures to show when they wrote down the same idea or have the same wondering. You might suggest they come up with a gesture that shows if they disagree with an observation (like they saw something different) or when they get a new wondering based on what someone else said.



Record observations and questions. After students have shared with their table group, call the class back together to share their noticings and wonderings. As shown on **slide J**, use a piece of chart paper or a shared digital space to record students' ideas as they share with the whole class.

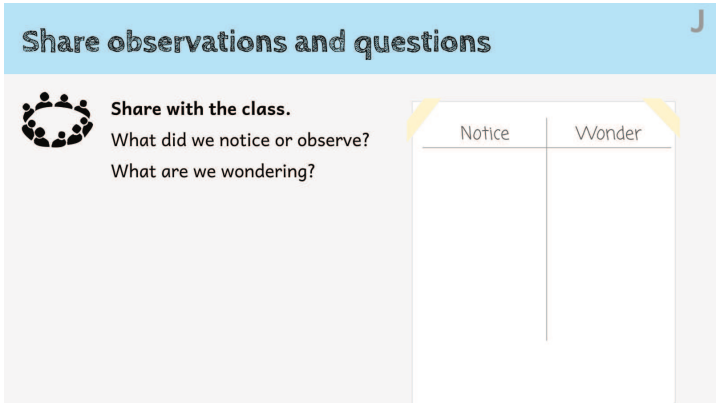
If students are unsure about sharing their noticings, you could prompt more specifically with questions such as

- What do you notice the different pictures have in common? (forest locations, fallen trees, things growing on and around the fallen trees)
- Did you notice any differences between the trees shown in the pictures?
- What do you notice about things growing?

If students are hesitant to share their wonderings, encourage them by sharing a couple of your own. Invite all student questions without critique, and avoid answering questions or offering explanations at this time. Again, the goal is to capture as many student-generated questions at this point as possible and to encourage students to think of new questions.

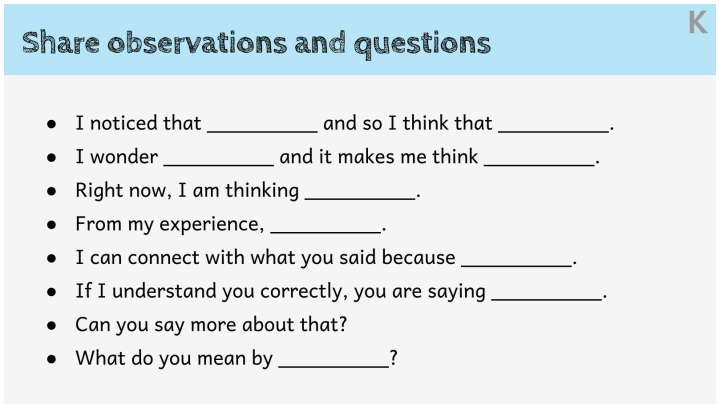
Broadening Access

The Notice and Wonder chart should capture not only students' ideas, but also the rich ways that students express their ideas. This is especially important for multilingual students because their language resources and practices are not always noticed or valued in school spaces. If a student shares an idea using words or phrases in a named language other than English (e.g., in Spanish, Arabic, Mandarin), record their idea exactly as they shared it and then add a translation in English next to it. If possible, have students record ideas onto the chart themselves. .



Prepare for an Initial Ideas Discussion. Use students' noticings and wonderings to transition to an Initial Ideas Discussion about the phenomenon we are observing.

- Explain to students that our goal in an Initial Ideas Discussion is to hear and see many of our early or first ideas. **Initial** means *beginning*, so these are ideas we are starting with, based on our observations of the different nurse logs.
- Point to the sentence starters on **slide K** that students could use during their discussion.



Teaching Tip

The purpose of an Initial Ideas Discussion is to provide an opportunity for students to share and make sense of their observations and ideas, starting toward an explanation even while the ideas are new and tentative or uncertain. Use talk moves like *revoicing* and *Say more* to help clarify the ideas so that others can work with them. The goal is to surface multiple ideas, so avoid privileging some ideas over others. Draw out possible competing ideas to motivate students at the end of the lesson to propose investigation ideas for those areas of uncertainty..

Facilitate an Initial Ideas Discussion. Consider saying something like, *Let's take a closer look at the pictures as we discuss our initial ideas about what we think is going on here.* Then use the following prompts to facilitate a class discussion about some of the details shown in the pictures and video:

Prompts to use	Ideas to look and listen for
Let's see if we can get any clues about what is going on here by thinking about the size of the different things we noticed. How big is the log/fallen tree? How can you tell?	*The log is huge! I can tell because of the way the person taking the video zoomed out and moved the camera to show the tree. One of the pictures also had people next to it, and they looked very small.*
You're right–the tree must have been very tall when it was standing! Based on what you know or have experienced, why does a tree usually fall in the forest in the first place?	*It might be dead.* *It could have gotten sick.* *A beaver may have chewed it down to build a house.*
We also noticed a lot of things that seem to be living and growing on the fallen tree–things that weren't there when the tree first fell, like we saw in the video. How about the other things that are on the log?	*They are much smaller.* *There are a lot of things on the log, which are much smaller than the log itself.*
Why do you think those things are on the log instead of next to it or somewhere else?	*The log must be giving something to, or helping, the other plants and things on it.*

Introduce the scientific name for the phenomenon. As on **slide L**, consider saying something like, *We get to spend some time thinking about the phenomenon we are seeing in these pictures, and it will be easier to talk about what's happening here if we have a name for these trees that fall in the forest. Scientists call* fallen trees that support other things to live and grow ***nurse logs***. *As we move through the unit we can see if we want to add to or revise this definition.*

Broadening Access

It is not important to identify the tree (or other things in the pictures) as living or nonliving. These characterizations depend on cultural perspectives. (For example, Indigenous cultures focus on the interconnected relationships among all things, so youth from Indigenous backgrounds may identify more things as living in comparison to more binary Western perspectives). Additionally, nurse logs contain more biomass than when they were standing trees, so there are many living and growing aspects to this once-living tree. Instead, focus students on observable characteristics, such as size, color, shape, and texture.

Literacy Supports

Eliciting students' ideas about related and familiar phrases for nurse logs provides an opportunity to engage in L5.5, as students demonstrate understanding of figurative language, word relationships, and nuances in word meanings.

Prompts to use	Ideas to look and listen for
Why do you think scientists use these words to describe what we have seen so far?	*Accept all student ideas.*
What does the word *nurse* mean to you?	*I think of my auntie who is a nurse.* *When my mom feeds my baby brother from her body, that's called nursing him.*
What do nurses do for people?	*They help people–at home, at doctor's offices, or in hospitals!* *So nurses help people, and nurse logs help things in a forest.*

Scientists' name for the phenomenon

L

It will be helpful in this unit to have a name for the phenomenon we are observing.

We noticed that the *fallen trees (logs) support other things to live and grow*.

Scientists call them **nurse logs**.

- Why do you think scientists use these words?
- What does the word nurse mean to you?
- What do nurses do for people?

As we move through the unit we can see if we want to add to or revise this definition.

SYNTHESIZE

Purpose: Emphasize that attempting to make sense of a phenomenon (in this case, how a nurse log helps other things live and grow) helps students uncover what they don't know for sure (so they can work to figure out those uncertainties).

2. Develop a model. [30 min]

Briefly define models. Introduce the idea of modeling to explain what we think might be going on with the nurse logs. Use **slide M** to explain what we mean by modeling in science class. The scientific process of modeling is not simply making a drawing or diagram; modeling helps us explain how or why something in the world happens.

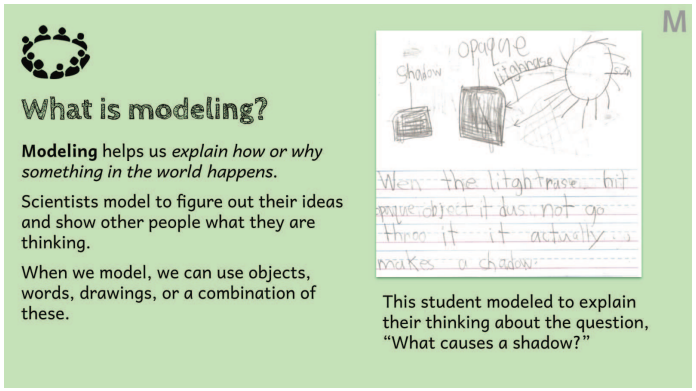
ASSESSMENT OPPORTUNITY

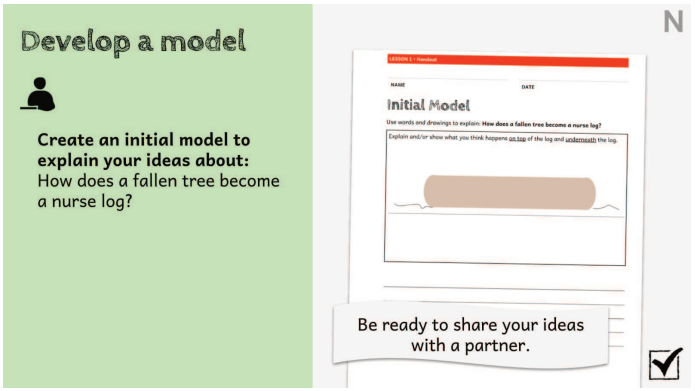
Pre-assessment: Students' initial models provide an opportunity to gather evidence around Learning Goal **1** with the purpose of determining support that students may need in upcoming lessons around the practice of modeling, and around ideas of scale, matter, and energy. Students will continue to develop these ideas and practices throughout this unit. Accept all student ideas. Refer to the Lesson Assessment Guidance at the beginning of the lesson.

Model our initial ideas about nurse logs. Display **slide N.** Distribute the Initial Model handout to each student. Review the prompts and give students about 10 minutes to model on the Initial Model handout.

Developing and Using Models

This is the first time students model to describe and predict a phenomenon. They will advance in this element of Developing and Using Models throughout this unit when they revise these models and create new models at different scales ranging from particles to ecosystems.





The prompts on the *Initial Model* handout ask students to consider what may be happening underneath and inside the log. You may need to reassure students that it's okay to not be able to explain exactly what is going on right now--scientists often model to help them understand what they haven't figured out yet! One way to reassure students about their uncertainty is to prompt them to add question marks to the places in their model where they're not sure how/whether something happens.

Form a Scientists Circle. Display **slide O.** Gather students in a circle (in chairs, sitting on a rug, or standing) in view of the chart paper and have them bring their completed *Initial Model* handout and something to write with. Students will remain in the circle for several minutes, although they will move among partner, small group, and whole class discussions. This will help minimize movement during class.

If your class does not often gather in a circle, explain a bit about why this structure is important to our work in science; students will be sharing their ideas with one another and to do that well they need to be able to hear and see everyone (not only the teacher).

Community Connections

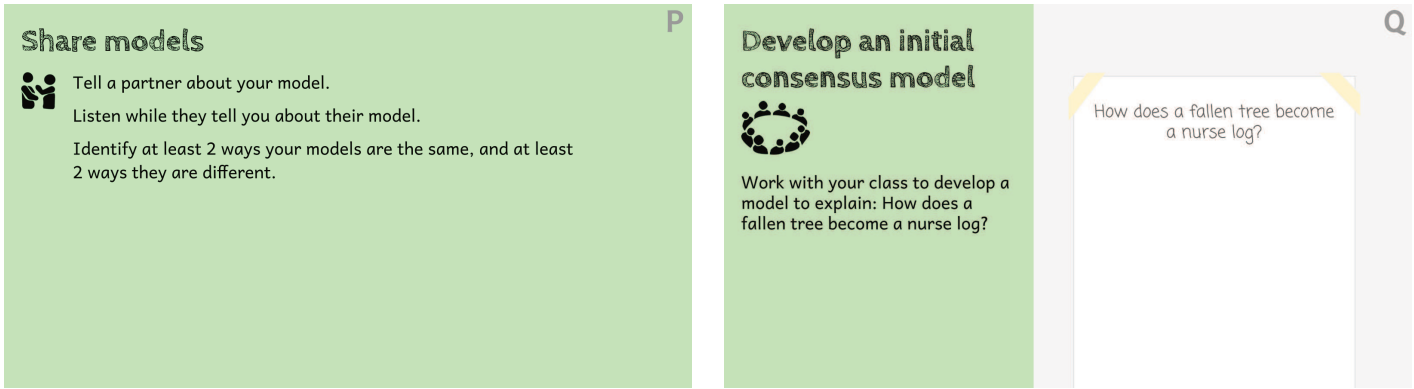
If this is your first science unit of the year, plan to spend about 30 minutes building classroom agreements using the Establishing Classroom Agreements teacher reference for how we figure things out in science, before engaging in the next component of this lesson. Establishing agreements for your classroom community now will allow students to practice them during the upcoming discussion.

If you have already established classroom agreements for science, take this opportunity to review them with students and add examples of how the agreements look, sound, and feel in your class.



Share initial models with a partner. As described on **slide P**, prompt students to turn to a neighbor and use their model to explain their ideas about how a fallen tree becomes a nurse log. After 1-2 minutes, prompt students to switch so that each partner shares their ideas.

Facilitate a Consensus Discussion. Display **slide Q.** Title a piece of chart paper or a shared digital space with the question we're trying to explain: "How does a fallen tree become a nurse log?"



Facilitate a Consensus Discussion to co-construct an initial consensus model including both views of the nurse log. Explain that in this case, consensus does not mean we all agree. We are trying to find if we do agree on some ideas, and to find places where we're not sure (because that will help us plan what we need to figure out!).

Broadening Access

Sitting (or standing) in a circle where they can face one another can help support students' engagement and build a sense of shared mission. The Scientists Circle is an important tool for developing students' agency in their learning as they take stock of what they have figured out and decide where they need to go next. Share with students that professional scientists also collaborate with one another to brainstorm, discuss, and review their work.

Literacy Supports

Engaging in the Consensus Discussion to build their initial consensus model connects students with the Classroom Agreements and provides an opportunity for students to develop and follow agreed-upon rules for discussion. This supports SL.51B as students follow these rules for discussions and carry out assigned roles.

Example prompts to start the Consensus Discussion

- Let's start by capturing everything you saw on or around the log.
- How should we model our ideas right now about how a fallen tree could become a nurse log?

To facilitate the Consensus Discussion, we recommend the following process:

1. Ask a student pair what their models had in common.
2. When they share an idea, ask the class if others had the same idea in their models. (You might invite students to use a hand signal so you can see responses from the whole class.)
    a. If many students included the idea, add that idea to the class model that you are building on the chart paper, asking the class to participate in or guide your drawing and labeling.
    b. When there is disagreement around an idea, add it to the class model with a question mark near it to indicate that this is something we are not sure about.
3. Continue to ask pairs what their models had in common or what was different, and add these ideas to the class model. Remember that the initial consensus model is intended to surface the range of ideas that students have, whether or not they are scientifically accurate.
4. If there are areas of uncertainty, write a question mark on the appropriate place on the model.

Direct students to use the sentence starters/questions on **Slide R** as they discuss:

Develop an initial consensus model R

- I think _____ because _____.
- I agree/disagree with that idea because _____.
- That idea makes sense to me because _____.
- I would like to add on _____.
- I am not sure yet about _____. Can you help me understand?
- But what about _____?
- I think you are saying _____. Am I right or can you clarify?
- It seems like we all agree that _____.

Broadening Access

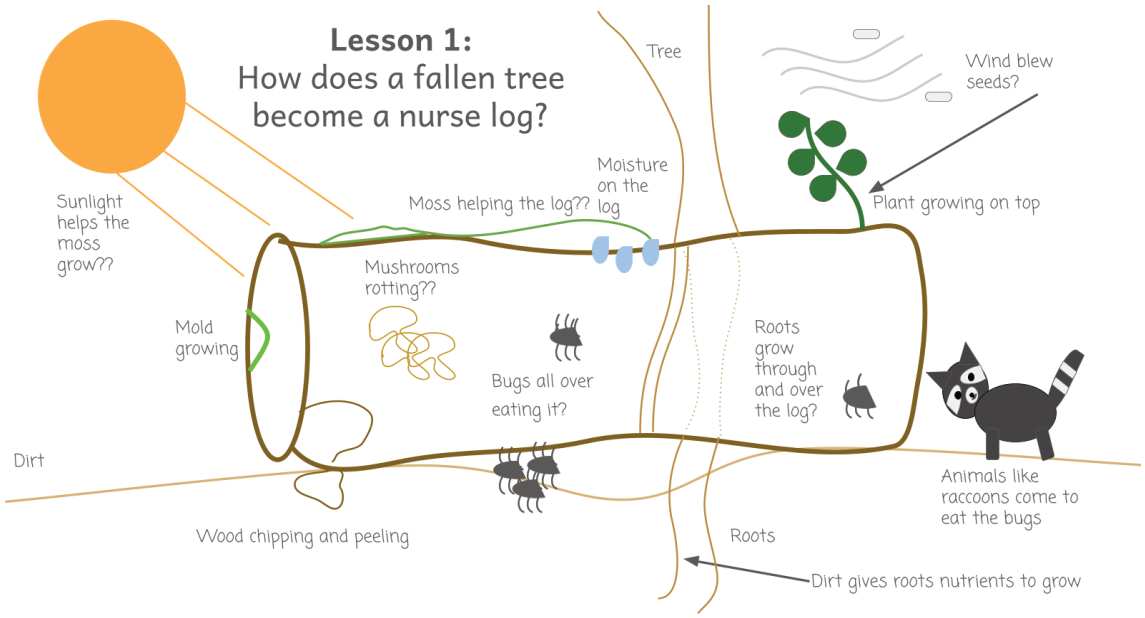
This Consensus Discussion provides an authentic opportunity for you to enhance students' language learning and language use for sensemaking work. You might find it helpful to use the Discussion Type Prompts teacher reference during the discussion. This handout provides teacher prompts that you could use to facilitate students in figuring out where we agree and where we disagree about how a fallen tree becomes a nurse log.

As the Consensus Discussion progresses, use the following questions to probe the parts of students' models that come from predicting what we would expect to find in, on, or under the nurse log (rather than directly observing).

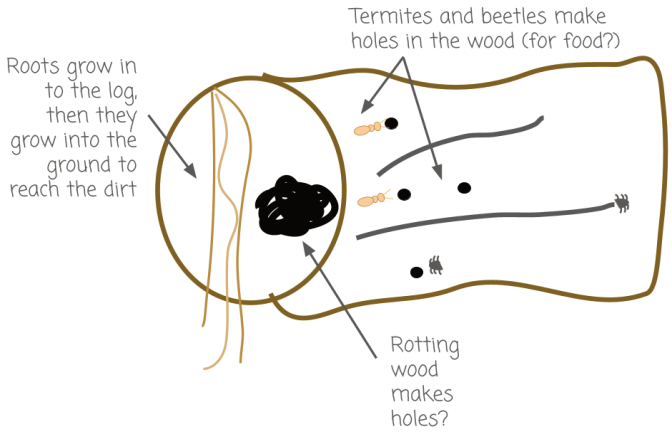
Prompts to use	**Ideas to look and listen for**
What do you notice on or near the log? What do you think the weather would feel like if you were near the log?	*Accept all student ideas. They may include (but do not need to correctly name or identify):* *We see plants, ferns, moss, mushrooms/fungi on the log. Near the log, we see dirt, fallen leaves, pine needles, other plants, trees, and ferns growing.* *I think it would feel warm and kind of sunny near the log. I can kind of see the Sun in some of the pictures and it's sunny in the video. The picture that has people in it shows them wearing shorts and T-shirts, so it is probably warm.*
What do you think we would find inside the log? What do you think it is like inside the log?	*Accept all student ideas. They may include (but do not need to correctly name or identify):* *We think there might be holes where little bugs are living inside of the log.* *Since there is a tree and plants growing on top of the log, I think there would also be roots from those plants growing into the nurse log.* *It's probably dark inside the log. It might get some water inside when it rains.*
What do you think might be under the log?	*Accept all student ideas. They may include (but do not need to correctly name or identify):*

What do you think it is like under the log?	*There might be animals like millipedes and roly-polies under the log.* *One time I found a salamander under a log! It was really damp under there.* *I bet the roots from the plants grow all the way through the nurse log and down into the soil, since plants need soil to grow.*

Example initial consensus models are shown here, but follow the thinking and ideas of the students in your class:



Lesson 1:
What if we cut the nurse log in half to see the inside?

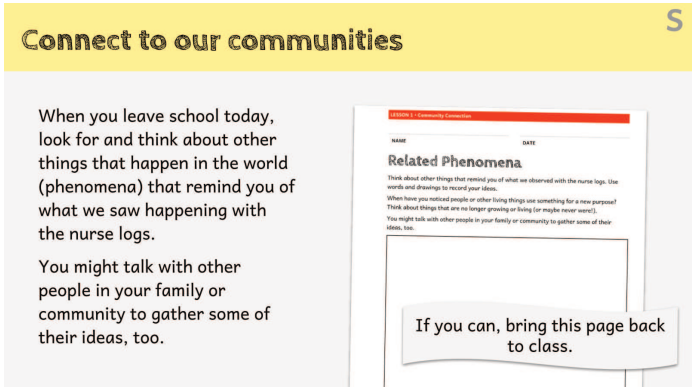


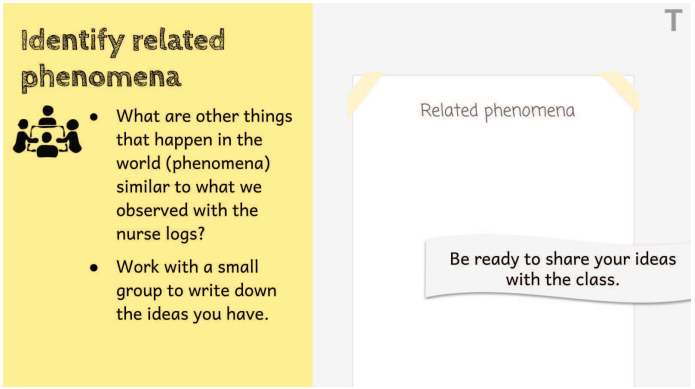
CONNECT

Purpose: Support students in identifying other phenomena like nurse logs to connect this science work to their lives and broaden what we figure out in this unit to explain more than a single phenomenon.

3. Identify related phenomena. [20 min]

Introduce the idea of related phenomena. Display **slide S**. Explain why we'll spend time thinking of other phenomena similar (or related) to what we observed about the nurse logs: when we figure out one phenomenon, that can help us explain others like it.





Invite students to identify related phenomena. Distribute the Related Phenomena handout to students and offer the opportunity to complete it outside of school before the next part of the lesson.

Identify related phenomena in groups. Display **slide T**:

- Form groups of 3-4 students.
- Give each group a large piece of paper and at least one marker (or, have them work in the same digital space).
- Give students 5-6 minutes to list examples of things that reminded them of what we saw happening with the nurse logs.
    - As on the *Related Phenomena* handout, guiding questions to prompt students' ideas include: *When have you noticed people or other living things use something for a new purpose? Think about things that are no longer growing or living (or maybe never were!).*

Teaching Tip

If only some students complete the *Related Phenomena* handout, that is fine. It should not be collected for a grade or score. Everyone will be able to participate in the Lesson 2 discussion whether or not they did this activity outside of school. Consider having some examples in mind that students may have encountered outside the classroom, and be prepared to talk about them with students during the Connect at the beginning of Lesson 2.

Share related phenomena. Have each student in the class briefly tell one example from their group's list and add these to a shared class chart. For phenomena that students share that don't seem related, ask students something like (without challenging them, coming from a place of curiosity): *Can you explain how you see those two phenomena as similar?*

NAVIGATE

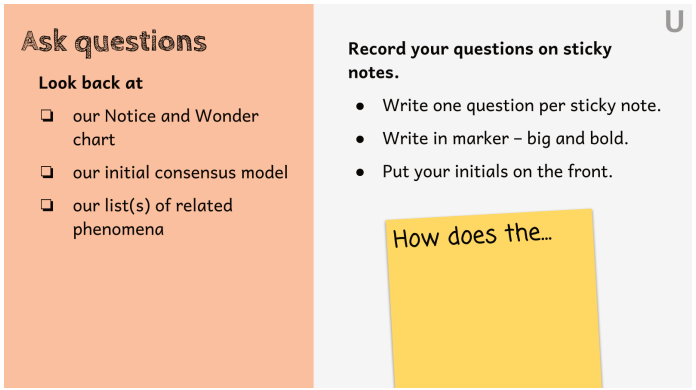
Purpose: Elicit students' questions about nurse logs to identify what the class wants to figure out throughout the unit and to emphasize the importance of students' questions driving the work.

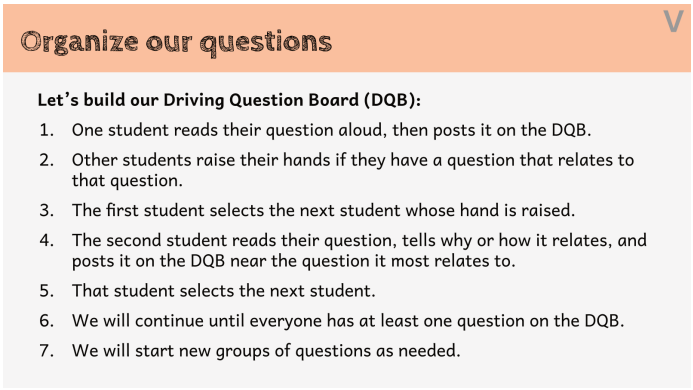
4. Ask questions. [30 min]

Write our questions about nurse logs. Display **slide U.**

- Distribute one sticky note to each student (or two if you have extra time or a small class).
- Direct students to look back at the work we have done thinking about **how the fallen tree becomes a nurse log**.
- Ask students to use a marker to write one question on their sticky note. They should write their questions so they are big and bold—we want to be able to see the questions clearly. Remind students that it is part of our mission in this unit to answer these questions.

Organize questions into a Driving Question Board (DQB). Gather students in a circle where they can see a piece of chart paper or other board space and have them bring their sticky-note questions. Follow the instructions on **slide V** to create your class Driving Question Board.

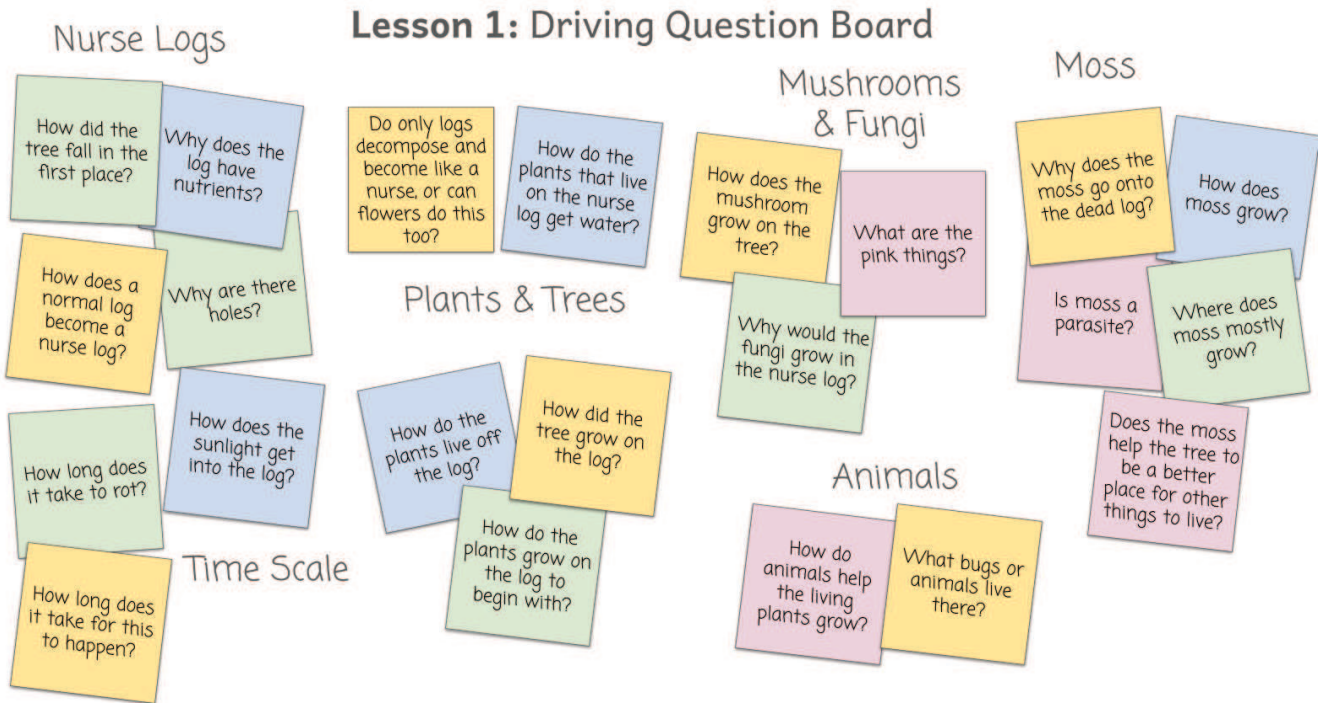




If you need to modify the DQB creation process based on your class size and culture, please remember the important features of this activity that you do not want to lose. This activity is intended to

- Make our questions public.
- Give us a chance to hear each other's questions.
- Move from 'my' question to a set of class questions to investigate– a class mission.
- Cluster similar-type questions.
- Give those clusters of similar questions a label we can quickly reference.

An example Driving Question Board is shown here, but follow the questions of the students in your class:



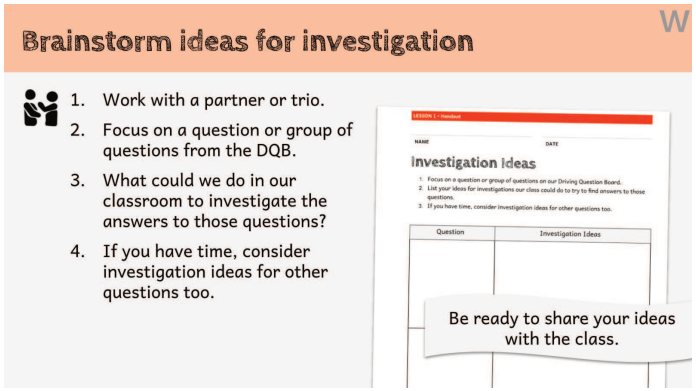
Identify the driving question for the unit. Ask students to propose an "umbrella" question that is big enough to include all the different kinds of questions we have. For example, "How could something that's not growing anymore help other things live and grow?" or "How does a nurse log help other things live and grow?" However, use language to honor what your students are thinking while also being similar to these examples. Write this unit question, or a similar question that your class develops, at the top of the DQB.

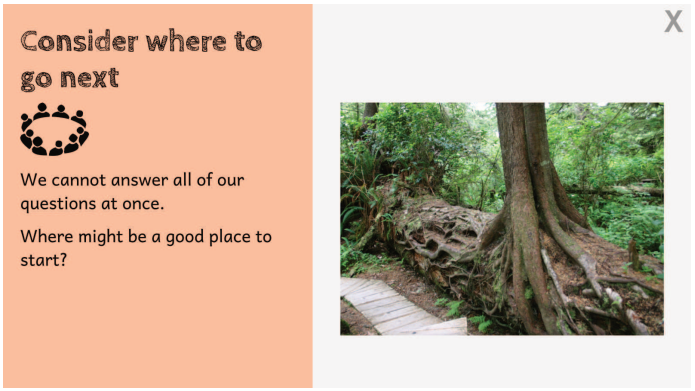
Brainstorm ideas for investigation. Display **slide W**. Have students find a partner or form groups of three with students they haven't worked with yet. Distribute the Investigation Ideas handout to each group. Give students about 5 minutes to generate investigation ideas.

If time allows, have groups share ideas with the whole class. You may wish to collect these ideas for investigation on a piece of chart paper or shared digital space.

Decide/consider where to go next. Display **slide X.** As you can, use students' questions from the DQB and ideas for investigation to narrow in on taking a closer look at the nurse log. Consider saying something like, *Gosh it was hard to see everything in those pictures! It would be great if we could see the nurse logs more up close. Let's start there next time.*

If you have not already done so, send home the Welcome Letter to introduce families to this unit.





Unit Advanced Preparation

URGENT: This unit involves live organisms.

- **PLEASE OPEN BOXES WITH LIVING ORGANISMS IMMEDIATELY.** While some of the materials can stay as they were sent for several days to a week, some of the organisms need immediate attention and have been shipped overnight to accommodate you (and to keep them alive!).
- All of the organisms chosen for this unit are very hardy but may experience mortality events in transit. If this happens, contact the vendor for your live organisms.
- **Note that the pillbugs/sowbugs are used in Lesson 9. You will need to unbox them upon arrival and set them up in their own terrarium separate from the Nurse Log in a Box kit that you will set up for Lesson 2.**
- When your living organisms arrive, unpack and care for them according to the care sheets included from the vendor. See the Guidance for Living Organisms teacher reference for more details on setting up and caring for the live organisms.

The following table identifies the lessons in this unit that require more significant advanced preparation. See the detailed preparation guidance as linked for each lesson.

Lesson and Lab	Approx. Total Prep Time	Preparation Overview	Preparation Details	Other
2: Explore Nurse Log in a Box	60	Set up 1 or 2 Nurse Log in a Box stations for students to observe the parts and organisms more closely.	Nurse Log & Moss Stations Setup	**When planning terrariums to use for your Nurse Log in a Box kit(s), be sure to keep the isopods (pillbugs/sowbugs) in a separate terrarium from the Nurse Log in a Box kits. If you combine them, the isopods will be too difficult to find again when needed for the Lesson 9 investigation.**
2: Observe Moss	30	Prepare moss samples for students to observe in small groups.		

After Lesson 2: (Teacher) Grow Mushrooms from a Kit	20 (including an overnight step)	Grow mushrooms for students to observe, including the block of organic matter from which they grow.	Guidance for Living Organisms	Start growing the mushrooms any time after Lesson 2, to be observed by students in Lesson 12. **In humid climates, the mushroom block may start growing in the bag in which it arrives. Consider starting the mushrooms within 1-2 weeks after receiving it.** As the first setup step, the mushroom block needs to be soaked in water for 6-10 hours to increase moisture. Then it takes 5-8 days for mushrooms to start growing, for a total of 10-12 days.
3: Giant Pumpkin Contest	45	Purchase 3 large fruits or vegetables of the same type--ideally medium-sized pumpkins, but any in-season produce that is irregularly shaped will do. In addition to the digital scale, gather any available measuring tools (rulers, measuring tapes, string, etc.).	Lesson 3 Materials and Preparation section	If you are not using pumpkins, consider changing the language on the slides for the investigation and on the Investigate How to Fairly Measure Pumpkins handout. Check the digital scale and other measurement tools to confirm that they can measure the pumpkins or other fruits/vegetables.
5: Investigate Matter	30	Prepare a bin or tray for each group of 3-4 students containing an empty paper cup, a cup of water, a flashlight, and a partially inflated basketball or sports ball. Be prepared to provide access to a digital scale.	Lesson 5 Materials and Preparation section	Any sports ball that needs a needle to inflate it will work for this investigation; you might borrow volleyballs, soccer balls, kickballs, etc. from a PE teacher or recess bin. ***Do not replace the sports ball with a balloon or beach ball***, as the change in volume between deflated and inflated will make it very difficult to observe any change in weight from added air.

6: Investigate Sunlight	30	Each group of 3-4 students will need one solar toy. Test the solar toys in the outdoor or sunny area where you are planning to have students conduct the investigation. If outdoors/sunlight is not an option, set up a heat lamp for students to use instead. Be prepared to show the Solar Toy Moving in the Sun video to help students connect to sunlight.	Lesson 6 Materials and Preparation section	
9: Investigate Isopods	45	Prepare isopods (pillbugs/sowbugs) for a heat transfer investigation.	Setup and Guidance for Isopod Investigation	To limit the number of live animals and materials needed, this investigation has been designed as a demonstration. If isopods are readily available in your area and you have access to more than one heat lamp, consider having students carry out the investigation in small groups. Alternatively, use the Isopod Energy Investigation video instead of conducting the investigation live.
12: Investigate Mushrooms	45	Prepare for a Jigsaw exploration where students explore mushrooms grown from a kit and images of mushrooms grown in light and dark environments.	Investigate Mushrooms Lab Setup and Guidance, Data and Instructions for Exploring Mushrooms	If you have multiple sections of students, you may need to purchase mushrooms from a store to use in the exploration, as some of the mushrooms grown from the kit need to remain attached to the block of substrate in the box.
14: Matter and Energy Flow Game	30	Prepare clips or a necklace for each student to display their game card in a visible place on their body. Review the game setup and play.	Game Guidance	

Guidance for Living Organisms

- **PLEASE OPEN BOXES WITH LIVING ORGANISMS IMMEDIATELY.** While some of the materials can stay as they were sent for several days to a week, some of the organisms need immediate attention and have been shipped overnight to accommodate you (and to keep them alive!).
- All of the organisms chosen for this unit are very hardy but may experience mortality events in transit. If this happens, contact the vendor for your live organisms.
- **Note that the pillbugs/sowbugs are used in Lesson 9. You will need to unbox them upon arrival and set them up in their own terrarium separate from the Nurse Log in a Box kit that you will set up for Lesson 2.**

Lesson 2: Moss



- Upon receiving the moss, poke holes in the containers they arrived in so they can drain and then place them on a plate or in another container (not included in your materials kit). Take care of the moss by spraying it with water every day or two until needed in students' investigations in Lesson 2.
- The moss is very hard to kill; it does not need to be kept in direct light.
- See the Nurse Log & Moss Stations Setup teacher reference for guidance on setting up the moss for students to observe.

Lesson 2: Nurse Log in a Box

- The Nurse Log in a Box will include several types of organisms like those listed below. What you receive in your kit may differ, depending on your vendor and the state in which you are located.
    - 5 adult mealworms
    - 5 pillbugs (separate from the other pillbugs needed for Lesson 9)
    - 2 millipedes
    - 2 bess beetles
    - 5 house cricket nymphs
    - 1 vegetative moss (separate from the other moss needed for Lesson 2)
- If you would like two Nurse Log in a Box stations (for easier viewing by students), you will need an additional terrarium and you can separate the materials provided into two terrariums. Do not add Nurse Log in a Box materials to the isopod habitat; they need a terrarium all their own so you will be able to find them again for the Lesson 9 investigation. See the Nurse Log & Moss Stations Setup teacher reference for guidance on how many stations to set up.

- Follow the directions provided by the materials vendor on how to set up the Nurse Log in a Box. It is ideal to first add the non-animal components (soil, wood, and moss/plants) and lightly spray to moisten, then let sit for about an hour before adding the invertebrates. You may do this using your gloved hands or a paper towel, or by putting the containers in the terrarium and letting the organisms crawl out (this takes longer).
    - See these two videos for an example of how to set up the Nurse Log in a Box:

Set Up the Nurse Log in a Box: Part 1, Set Up the Nurse Log in a Box: Part 2



- Once assembled, the kit is nearly self-sustaining!
- After setup, every 2-3 days
    - Lightly mist the wood, soil, and organisms with water to moisten.
    - You may add small slices of apple, potato, or carrot for the invertebrates.
    - Inspect and remove any food or other components before they rot.

Lesson 9: Pillbugs / Sowbugs (Terrestrial Isopods)

- Depending on availability from your materials vendor, you may receive either pillbugs or sowbugs.
    - If you live in an area that has isopods, consider collecting or having students collect additional isopods for student groups to conduct the investigation.
- Open the shipment immediately upon arrival and remove the container.
- Open the container and inspect the paper towel that is inside: most of the isopods will likely be found in the folds of the paper towel.
- Moisten the towel by sprinkling or spraying it with water. The isopods breathe with gills and need moisture for gas exchange. Place the towel with the isopods in a terrarium. **Do not add the isopods to the Nurse Log in a Box** terrarium or it will be challenging to gather them for the investigation.
- Add the soil that is in the container and dispose of the container.



- If possible, add a bit more soil, leaf litter, compost and/or mulch to the terrarium to cover the bottom (either from your local area or from the Nurse Log in a Box kit). **Do not use pine bark or other mulch that has been treated with chemicals.**
    - If soil or leaf litter is not accessible, this step is not necessary.
- Mist the container. If possible, leave the terrarium in a dark area for at least an hour.
- To maintain the isopods until they are needed in Lesson 9
    - Place the habitat in an area that receives indirect light and maintains a temperature of 20 to 22 °C (68 to 72 °F).
    - Every 1-3 days, lightly mist the terrarium and soil.
    - Replace the paper towel with another moistened towel if it shows signs of mold.
    - Otherwise, isopods require minimal care since they are detrivores. If you are unable to add leaf litter or compost to the terrarium, offer

additional food once a week, such as: fish food flakes, moistened dog food, leaf litter, or pieces of raw potato or apple. Immediately remove any material that shows signs of mold.

Lesson 12: Grow Your Own Mushrooms

- Start growing the mushrooms any time after Lesson 2, to be observed by students in Lesson 12. In humid climates, the mushroom block may start growing in the bag in which it arrives. Consider starting the mushrooms within 1-2 weeks after receiving it.
- As the first setup step, the mushroom block needs to be soaked in water for 6-10 hours to increase moisture. Then it takes 5-8 days for mushrooms to start growing, for a total of 10-12 days. Here is a time-lapse video of the growing process: Mushrooms Growing video.
- Follow the instructions included with the kit to set up and grow the mushrooms. Once they start growing, inspect and mist lightly every day or two until needed for Lesson 12. See the [EM.L12.TREF1, 1] teacher reference for more guidance on preparing for the lesson.
- You can also grow a second crop of mushrooms from the same kit! Watch this video to assist you: Second Crop of Mushrooms.

What to do with live organisms after investigations

- When the class has finished investigations involving the live organisms, you can continue to support the living organisms per the instructions provided by your materials vendor.
    - We strongly recommend giving any unwanted organisms to another individual. Some of the invertebrates may be fed to pet amphibians or reptiles!
    - We do not advocate the release of any of the fungi, plants, or animals into your environment, due to the possibility of disrupting the local ecosystem. Please contact your local (state) Department of Agriculture for any restrictions on the release of organisms.
    - As a last resort, place unwanted organisms in a sealed container in a freezer for 48 hours. Dispose of the organisms in the regular solid waste. The most humane way to euthanize any of the living organisms in this kit is through freezing temperatures.

Additional Safety Tips

- Use appropriate containers or terrariums for housing the invertebrates and other organisms. Ensure that the containers have proper ventilation to maintain air circulation and prevent the buildup of harmful gasses.
- Consider wearing non-latex gloves and safety glasses with side shields or safety goggles when handling invertebrates or their habitat materials, to prevent direct contact with any potential irritants or pathogens.
- Provide necessary safety equipment such as tweezers or small brushes for handling insects or other small animals without direct contact. This minimizes the risk of accidental injury to both the invertebrates and the students.
- Always directly supervise students when they are working with insects or their habitats.
- Have clear emergency procedures in place in case of accidents or invertebrate related incidents. Make sure students are instructed on what to do in case of stings, bites, or other injuries.

- Be aware of any students or staff members who may have allergies to insect stings/bites or other allergens associated with insect or other animal culturing such as mold and mildew. Take necessary precautions to minimize exposure and ensure a safe learning environment for everyone.
- Keep the habitats clean and well-maintained to prevent the buildup of mold, bacteria, or other harmful microorganisms. Regularly remove any waste or debris from the habitat to maintain hygiene.
- Dispose of any leftover food or organic waste from the habitat in a proper manner to prevent contamination and odors.
- Remind students to wash their hands thoroughly with soap and water after handling insects or their habitats.

Establishing Classroom Agreements

	duration	slides	materials
NAVIGATE			
1. **Introduce our purpose.** Connect the work of our current unit to today's work of drafting initial classroom agreements.	5 minutes	A	Classroom Agreements Slides
EXPLORE			
2. **Make observations.** Observe photos of people doing science in different ways; share with partners and discuss as a class how all of those people are doing science.	10 minutes	B-F	
SYNTHESIZE			
3. Participate in a discussion. Co-construct first-draft descriptors for what suggested classroom agreements mean for our class.	15 minutes	G-M	Classroom Agreements handout, Classroom Agreements for How We Figure Things Out in Science chart
Total time for this lesson: 30 minutes			

Lesson Materials and Preparation

Per student:	**Per partners:**	**Per small group:**	**Per class:**
• *Classroom Agreements* handout	• N/A	• N/A	• Classroom Agreements Slides • chart paper

Preparation Checklist • Estimated time: 10 minutes

- Be ready to use Classroom Agreements Slides.
- For more information about why classroom agreements are critical to students' science work and how you can incorporate them, take about 10 minutes to read the "Building a Classroom Community for Science" section of the Teacher Handbook.
- Title a piece of chart paper, "Classroom Agreements for How We Figure Things Out in Science". As you co-construct agreements and descriptions with the class, note that you will revisit and add to them throughout the year, so it will be helpful to leave space for additional ideas and examples where you describe each agreement. Throughout the year, plan to add examples of what an agreement means to your class and/or examples of students upholding the agreement, such as photos of students engaging in lessons, completed student handouts, or other student artifacts that the class feels capture the agreement.
- If you plan to have students keep a copy of the Classroom Agreements in their science notebooks (such as taped inside the front cover), consider building them in a digital space, such as by typing into a copy of the Classroom Agreements handout. Then you can print the page for students to add to their notebooks. Be sure to leave space for additional ideas that you will add throughout the year.
- As your class works to look, listen, and respond to one another's ideas, you and your students may find one or both of these resources helpful for supporting interactions among classroom members: Icons with Discussion Supports and Discussion Supports.

Classroom Agreements Vocabulary

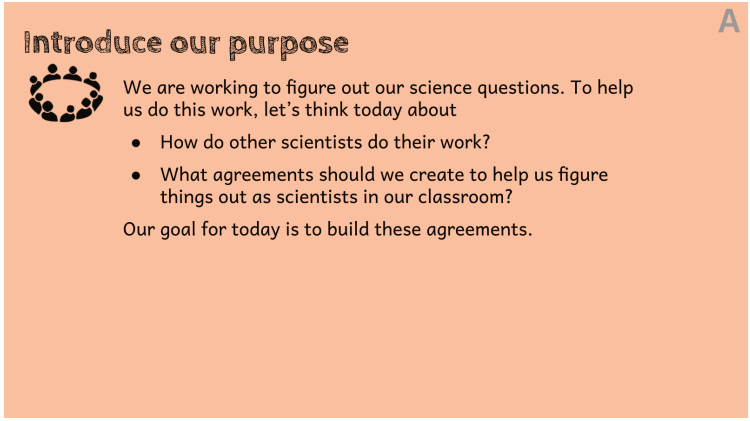
	NAVIGATE	EXPLORE	SYNTHESIZE
Science Practices and Concepts that Build over Time These are words best learned as students engage with them over time. We will revisit them across units and grade levels.	agreement		

NAVIGATE

Purpose: Connect to the unit work students are currently beginning to motivate the need for drafting initial classroom agreements. We will continue to revisit, revise, and add to these agreements throughout the unit and in future units.

1. Introduce our purpose. [5 min]

Motivate the need for classroom agreements in science. Gather students in a Scientists Circle so everyone can see and hear one another, and see the slide images and chart paper you've set up. Display **slide A**. Introduce the purpose of our work today by telling students that we are excited to be figuring out _____(insert unit anchor). Help students understand that the work of figuring things out in science requires individuals to work together in a community and use different ideas from different people. Since we are a community of individuals doing science, we should agree on how and why we want to work together to figure things out. Elicit students' ideas about the word "agreement" using the following questions.



Prompts to use	Ideas to look and listen for
What do you think "agreement" means?	*Accept all ideas.*
Do any of you have agreements with your family or community members?	*Accept all ideas, likely related to how we treat each other or uphold responsibilities.*
Why do you think scientists might make agreements?	*Accept all ideas.*

Summarize by defining an **agreement** as *an understanding we have between one another*. In this case, we want to understand how to help our classroom community figure things out together.

Teaching Tip

Throughout this lesson we say "science," but we fully expect these agreements to support our work as engineers, also. If your current unit includes engineering, consider interspersing the word "engineer" as you establish these agreements.

Teaching Tip

The scientific community relies on different perspectives and methods to get new ideas, clarify and build on ideas, and find consensus around what we agree on and what we still question. The agreements you co-develop with students support this integral aspect of science work: that it is necessary to work and learn from others' different ideas to figure things out together.

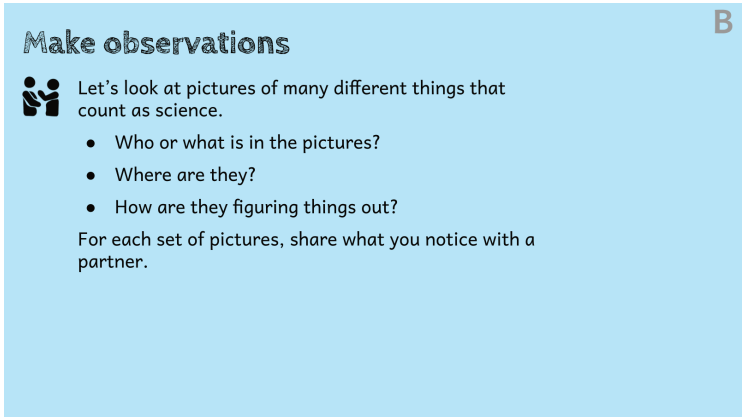
EXPLORE

Purpose: Guide students in noticing how science can be done by all kinds of people and in many different ways and places, and use those ideas to begin co-constructing our classroom agreements. We will continue to find and discuss more examples throughout the year as we engage in more units.

2. Make observations. [10 min]

Discuss images of people doing science. Display **slide B**. Explain that we will observe some images on the following slides that all count as science. For each set of images, have students share their thoughts with a partner about questions such as these:

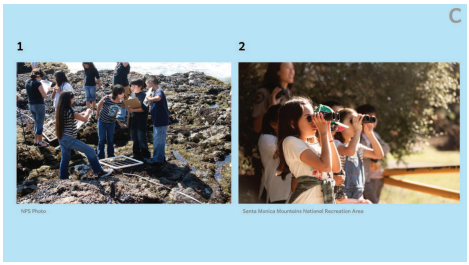
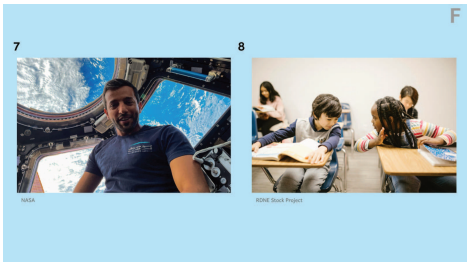
- Who or what is in the pictures?
- Where are they?
- How are they figuring things out? (What are they doing?)



Display **slides C–F**, pausing on each slide to invite students to talk with a partner again as they continue to observe the different ways and places people do science.

Teaching Tip

Because students will be sitting for the discussion in the next step, we suggest that you use an activity structure your students are familiar with, such as inside-outside circles, to allow and encourage them to move while they talk with partners about their observations.

Slide	Prompts to use	Ideas to look and listen for
	Who/what is in these pictures? Where are they? How are they figuring things out?	*students, kids, in groups* *outside, in a park* *observing, taking notes, using tools to look closely, wondering, talking*
	Who/what is in these pictures? Where are they? How are they figuring things out?	*a class of kids plus teacher, a kid working alone* *inside, in classrooms* *asking questions, reading, listening, looking at a chart, writing on a chart or notebook*
	Who/what is in these pictures? Where are they? How are they figuring things out?	*adult(s) and kid(s), plants, tools* *outside and inside* *planting, growing things, investigating*
	Who/what is in these pictures? Where are they? How are they figuring things out?	*an adult and kids* *in some kind of ship, in a classroom* *could be on a video call, maybe speaking a language other than English, talking, sharing ideas*

Navigate toward defining our classroom agreements by saying something like, *Let's think about all these different ways and places people do science as we build our agreements for doing science together.*

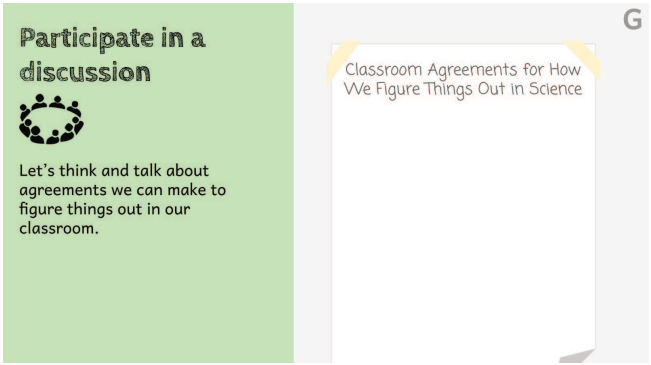
SYNTHESIZE

Purpose: Support students in describing how the agreements for figuring things out in science will look and sound in this classroom; emphasize that the class will continue to refine, expand upon, and add details and examples to these agreements as they work together as scientists this year.

3. Participate in a discussion. [15 min]

Explain how we will describe what these agreements mean to us. Display **slide G.** Title a piece of chart paper, "Classroom Agreements for How We Figure Things Out in Science". Distribute the Classroom Agreements handout for students to use as thinking space as we discuss each agreement.

Explain that as we list classroom agreements, we will describe how we can help each other with the agreements and what they will look like and sound like in our classroom.





Write our first classroom agreement. Refer back to the images collected on **slide H** and the ideas students shared with partners about who, where, and how we do science. Help students notice the variety of people, places, and ways we can work to figure things out. Consider asking something like, *Can we agree that science can be done in many different ways?* Elicit yes.

Add this idea to the chart as your first classroom agreement, adjusting the wording as needed to best align with how your students were expressing it.

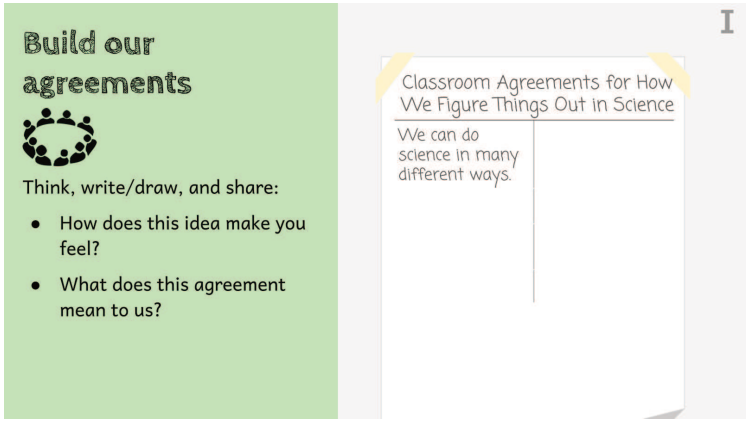
Teaching Tip

Students do not need to add this handout to their science notebooks. You will capture the agreed-upon descriptors in a public classroom space and, if you wish, print copies for students to add to their notebooks.

Teaching Tip

Suggested agreements were chosen intentionally and co-constructed with K-5 teachers from across the country, but it is ultimately more important that they reflect your classroom community's understandings and commitments to one another. Choose to express each agreement in the ways that make sense to your students (their words, gestures, connections to other agreements your class has, etc.) For ideas about how to make the agreements your own, see Teacher Handbook.

Check in on our feelings. Explain that we need to take a moment to consider how each of our agreements makes us feel (refer to **slide I**). If any agreements make us feel sad, shy, or scared, we can plan how we will help one another when we feel that way. Consider asking something like, *How does it make you feel that we can do science in many different ways?* Invite and validate all feelings students share.



Describe our first agreement. Ask something like, *What does this agreement mean to us? How will it look, sound, and feel in our classroom when we do science in many different ways?* Give students a minute or two to write or draw on their handout, then share ideas with the class.

As students share, ask for a hand signal or head nod if this descriptor is something we should include to help us know how these agreements will look and sound in our classroom. Capture ideas, examples, and descriptions for this agreement in your chart using the students' own words and/or drawings. See the sample chart at the end of this lesson plan and other examples in the Teacher Handbook.

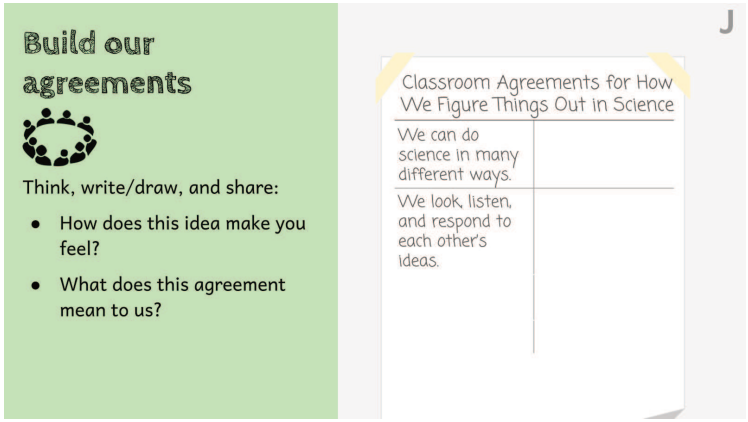
Write and discuss our second agreement. Point out that we also noticed scientists working together and sharing ideas in the pictures (refer back to the images collected on **slide H**). Propose an agreement about what we do when our fellow scientists in this classroom share their ideas. We should look, listen, and respond to one another's ideas (refer to **slide J**). Give students a minute or two to write or draw on their handout, then share ideas with the class.

Broadening Access

Allow students plenty of "think time," and expect them to share their ideas in many ways (e.g., through a drawing, across different languages). It is okay if students decide not to share their ideas; you might look over their shoulders while they write, draw, and/or talk to be sure everyone's ideas are captured.

Teaching Tip

If many students feel sad/scared/shy about an agreement, consider whether you will modify it and keep it, or remove it from your list. It is important to consider students' emotions because how they feel can positively or negatively impact their sensemaking work.



Prompts to use	Ideas to look and listen for
How does it make you feel that we will look, listen, and respond to each other's ideas?	*(Invite and validate all feelings students share.)*
Can we agree to look, listen, and respond to each other's ideas?	*Yes (or discuss any concerns/clarifications as needed).*
What does this agreement mean to us? What will we see, hear, and do in our classroom when we look, listen, and respond to each other's ideas?	*We ask each other, "What do you think?" and "What makes you think that?"* *We restate and re-represent each other's ideas.* *We say, "I agree with you because ____" or "I disagree with you because ____."*

Add this agreement and description to the chart, adjusting the wording as needed to best align with how your students were expressing these ideas.

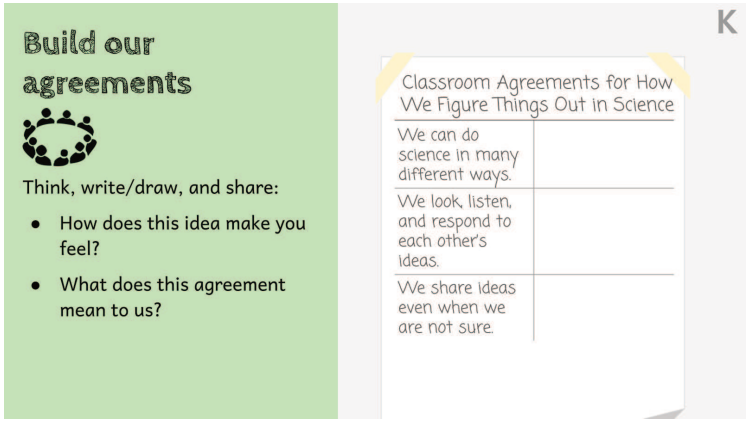
Teaching Tip

Often, teachers and students think that we listen to each other to find answers or check if we are wrong or right. Instead, encourage students (and yourself!) to look and listen to each other to compare how our thinking is similar or different and stretch our thinking in new ways.

Community Connections

As your class works this year to look, listen, and respond to one another's ideas, you may find one or both of these resources helpful: Icons with Discussion Supports and Discussion Supports. The hope is that over time, all classroom members feel comfortable engaging with each other in these ways and that they internalize these kinds of interactions and make them their own. These interactions will support the class's sensemaking work.

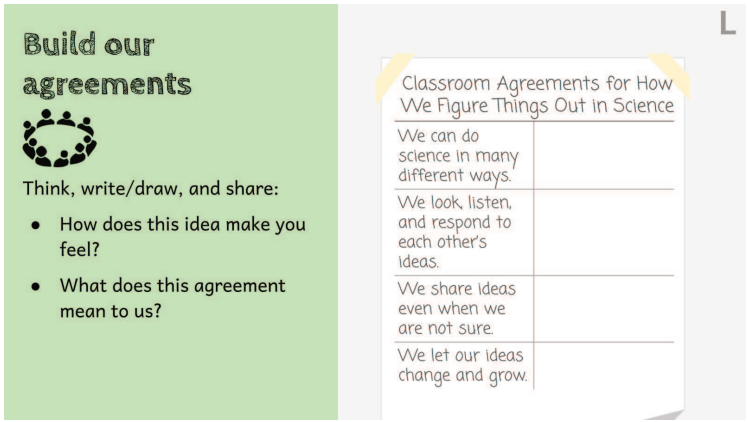
Write and discuss our third agreement. Point out that we thought people in the pictures were asking questions or wondering about things (refer back to the images collected on **slide H**). Remind students that scientists do work to figure things out, but they usually have more questions than answers. Science is all about wondering and trying things out. So, it is important for scientists to share their ideas even when they are not sure (refer to **slide K**).



Prompts to use	Ideas to look and listen for
How does it make you feel to share ideas even if you're not sure?	*(Invite and validate all feelings students share.)*
Can we agree to share ideas even when we are not sure?	*Yes (or discuss any concerns/clarifications as needed).*
What does this agreement mean to us? What will it look like and sound like to share ideas even when we are not sure?	*We encourage others to share their ideas.* *We say, "I'm not sure, but I think ____."*

Add this agreement and description to the chart, adjusting the wording as needed to best align with how your students were expressing these ideas.

Write and discuss our final agreement. Remind students that science is the work of figuring things out. We will have questions and try to find answers in many different ways. Connect to the work students have been doing already in your current unit. As we figure things out, our ideas will grow and change (refer to **slide L**).



Prompts to use	Ideas to look and listen for
How does it make you feel to let your ideas change and grow?	*(Invite and validate all feelings students share.)*
Can we agree to let our ideas change and grow?	*Yes (or discuss any concerns/clarifications as needed).*
What does this agreement mean to us? What will it look like and sound like to let our ideas change and grow?	*It is okay to change our minds when we find new evidence.* *We say, "I used to think _____, but now I think _____ because _____."* *We use and build on each other's ideas.*

Add this agreement and description to the chart, adjusting the wording as needed to best align with how your students were expressing these ideas.

Your class may also choose to create a fifth agreement in addition to the suggested four. You might ask, *What else do you need from our classroom community to feel valued here or to do science together?* It is important to provide space for students to add additional agreements that will help them best engage in sensemaking work; this can be done at various points of the year, especially as students begin to engage in science in new ways. Note that we do not need a classroom agreement about safety considerations because we will always have specific directions (from the teacher) about how to use materials safely—those directions are not up for discussion like these agreements are.

Elevate the purpose for working together to figure things out. Refer to the chart and point out that each of these agreements will help us share our ideas and build our ideas together, not alone. Ask something like, *Why is that important? Why does sharing ideas help us figure things out in science?*

Ideas to look and listen for

- We need to decide/agree together if we found an answer to a question.
- We can't find answers by ourselves.
- Engaging with different people's ideas will help us understand better (more than keeping to our own ideas).
- We have too many questions to answer alone.

Celebrate creating these agreements. Display **slide M**. Celebrate the work we did today to establish these agreements, and point out again that we are not finished– these agreements are "living" and will continue to grow and change as we use them to do this work. Tell students that we will continue to add pictures, examples, and ideas to this chart as we use our agreements and further describe what they mean to us.



Students might want to sign the chart to show their commitment.

Transition back to your next unit-specific component by encouraging students to start using these agreements to do our science work.

See the example below as a reference--your classroom's agreements will look different because you will use your students' ideas, words, and drawings. Your agreements might include statements, sentence starters, and references to other expectations/norms/rules that you already have in your classroom or school.

Classroom Agreements for How We Figure Things Out in Science	
Agreement	What does this agreement mean to us?
We can do science in many different ways.	• There is no one way to do science. • Everyone can do science. • We share our science ideas in many ways (words, gestures, graphs, etc.). • We use different tools to do science.
We look, listen, and respond to each other's ideas.	• We value each other's ideas. • We ask each other, "What do you think?" and "What makes you think that?". • We restate and re-represent each other's ideas. • We say, "I agree with you because ____" or "I disagree with you because _____."
We share ideas even when we are not sure.	• We encourage others to share their ideas. • We say, "I'm not sure, but I think _____." • We know mistakes help us learn.
We let our ideas change and grow.	• It is okay to change our minds. • We say, "I used to think _____, but now I think _____ because _____." • Our ideas might change or grow with new evidence. • We use and build on each other's ideas.
(What else do we need?)	

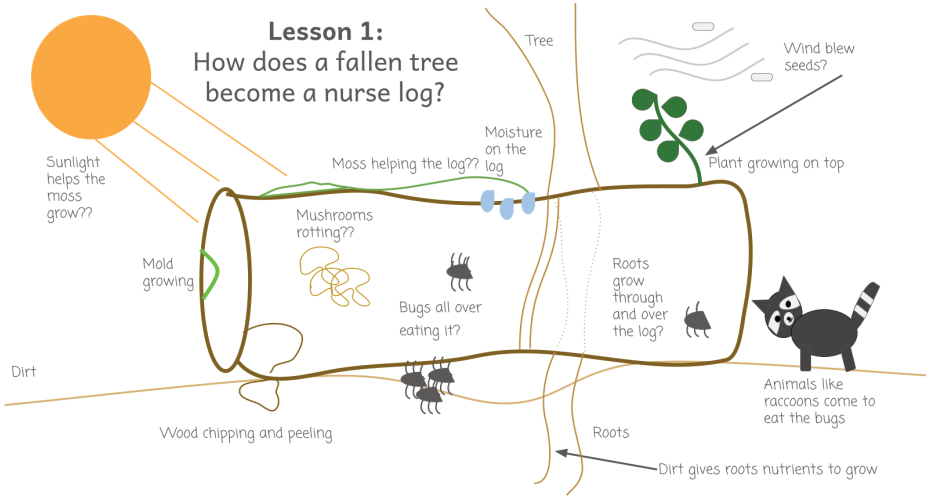
END OF CLASSROOM AGREEMENTS LESSON

Example Unit Models and Charts

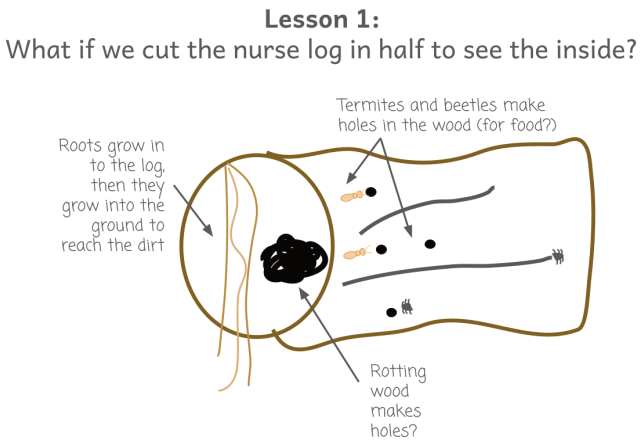
Lessons in this unit may have a custom chart to support the learning in that lesson only, but there are some charts that need to be available throughout the unit. For all charts, you may use chart paper, a whiteboard, or a digital space for your students to share their thinking. Use what is best for your classroom and your students' sensemaking; however, make sure that charts needed throughout the unit are easily accessible by your students.

The Driving Question Board (DQB) and Class Consensus Model will be used frequently throughout the unit and need to be readily available and viewable by students. These charts may be combined or co-created as separate charts. If you leave them on the wall for the duration of the unit, they should be centrally located on your wall space. Alternatively, they can be easily brought out for science class. Students will need access to them so they can add and answer questions throughout the unit, and add to and revise the model. The following images show examples of the initial DQB and key models that students develop throughout the unit. These are only suggestions for how students might represent components (including matter and energy) and interactions on their models.

Lesson 1: Initial Consensus Model

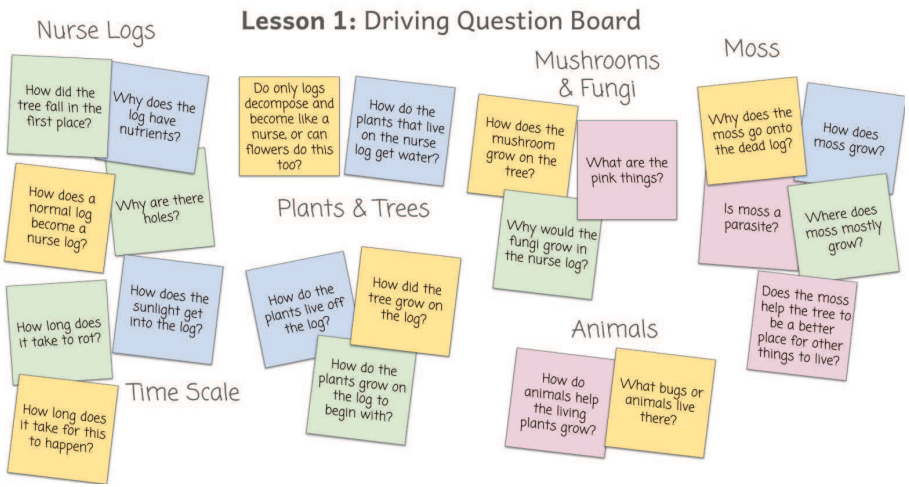


Lesson 1: Initial Consensus Model



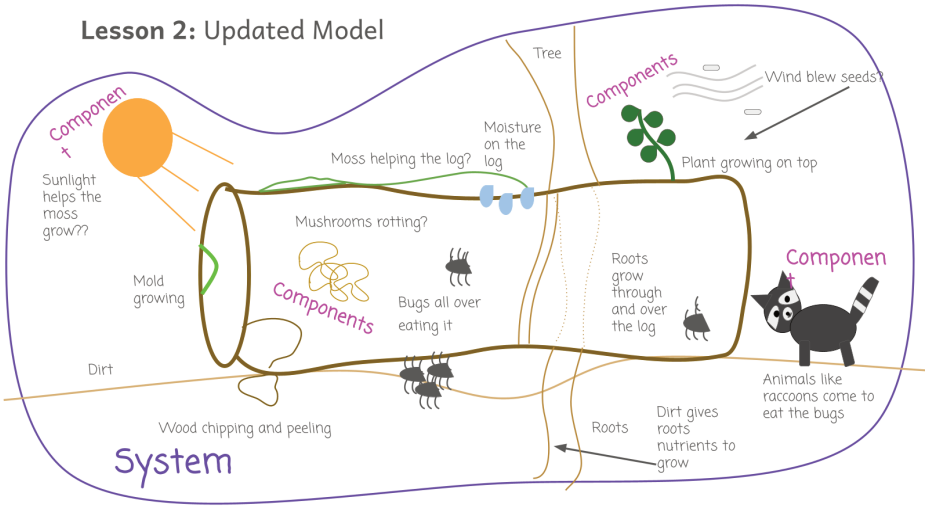
In Lesson 1, students individually model and then the class creates an initial consensus model to explain how a fallen tree becomes a nurse log. Students will likely include ideas about the different parts that they noticed, including plants, moss, and fungi, as well as parts that they did not directly observe, such as holes and animals. We suggest using arrows to point to specific parts of the model and dashed lines for parts that students suggest but that we cannot directly observe (like the roots growing through the log). For all modeling work in this unit, follow students' lead in writing and drawing on the model, adding question marks to indicate areas of disagreement or uncertainty.

Lesson 1: Driving Question Board



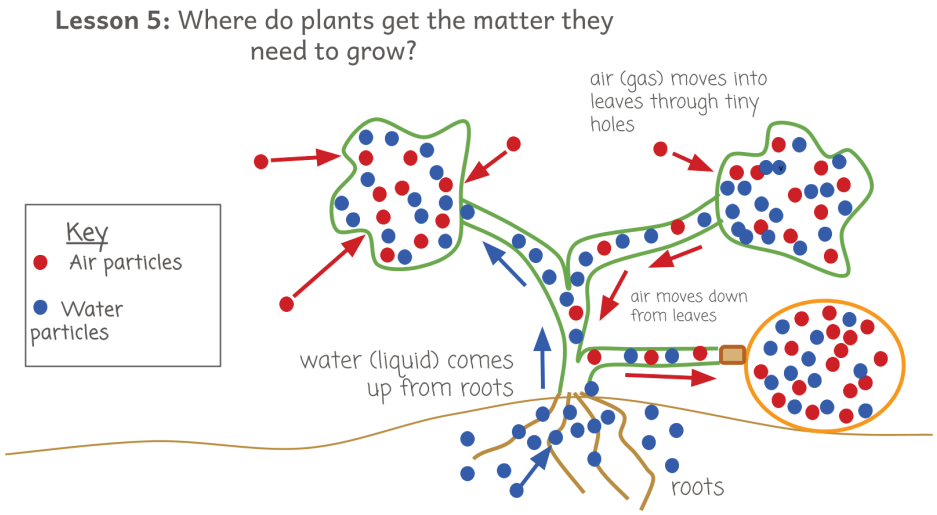
At the end of Lesson 1, the class creates a Driving Question Board of their wonderings that will guide the investigations in this unit. We have included an example of a DQB (based on real students' questions) that is separate from the model, but students could add their questions on sticky notes directly to the relevant part of the model. Construct your DQB in a way that is useful for your classroom (separate or in combination with the model).

Lesson 2: Updated Consensus Model

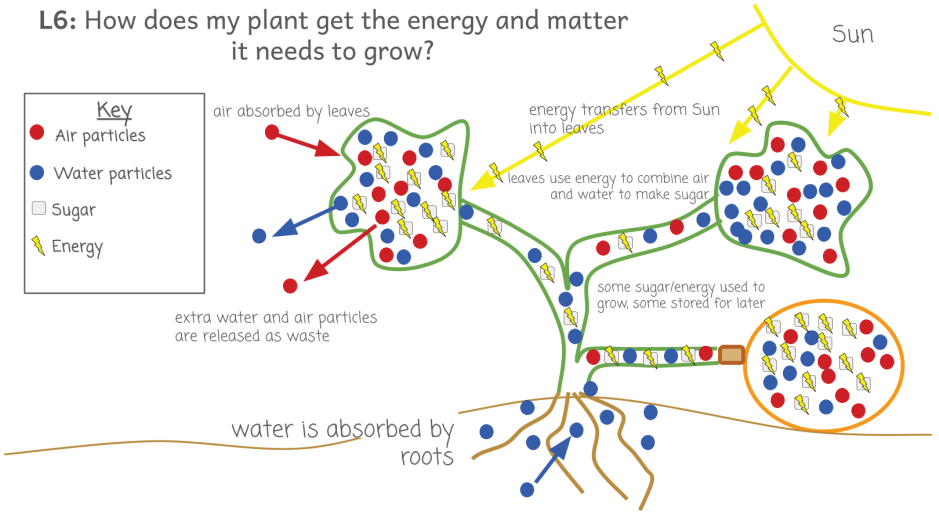


In Lesson 2, the class adds new parts to the initial consensus model based on their new observations of a Nurse Log in a Box, and pictures and videos that enable them to take a closer look. They further characterize the parts as components, and all of the nurse log components as a system since the components are working together.

Lesson 5: Source of Matter for Plants

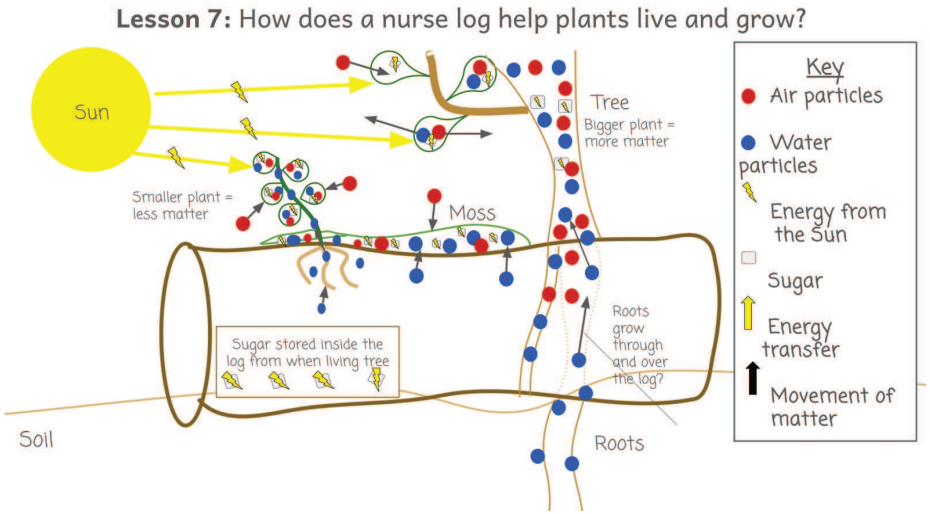


Lesson 6: Source of Energy for Plants

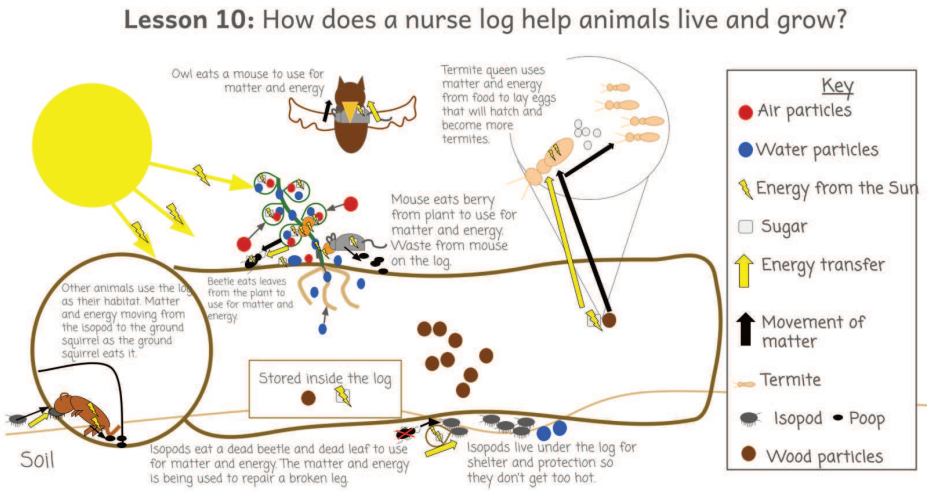


In Lessons 5 and 6, students model to explain where plants get matter and energy needed to grow. They leverage what they figure out about particles in Lesson 5 to represent air and water moving into the plant. They add on to the same model in Lesson 6 to represent how plants transfer energy from the Sun and use the energy to combine air and water to make sugar, which is used for growth and stored for later use. Each lesson provides suggested prompts for how to elicit and leverage students' ideas about how to represent these complex and abstract concepts.

Lesson 7: Updated Consensus Model

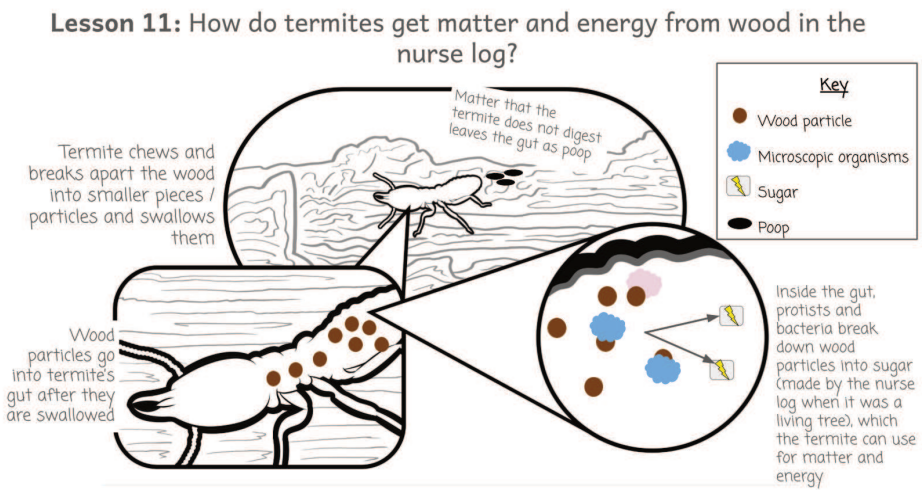


Lesson 10: Updated Consensus Model



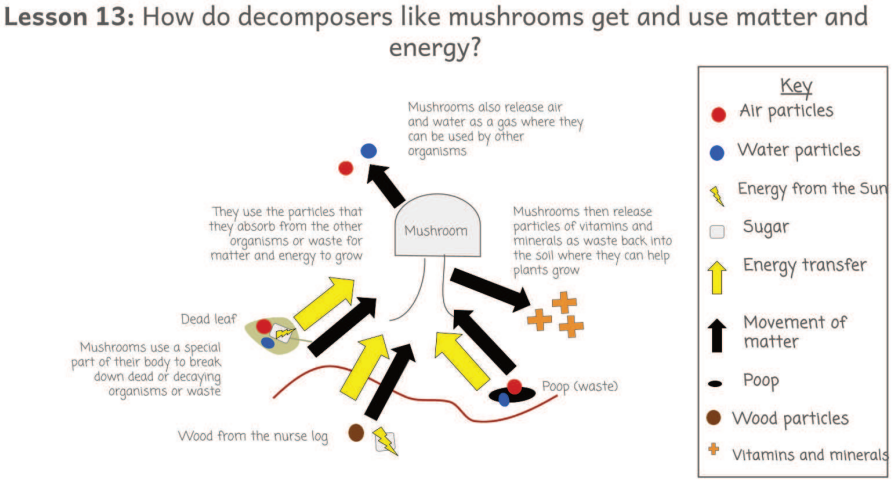
In Lesson 7, the class returns to the nurse log model to explain how the nurse log helps plants live and grow, which was a main category of questions on the DQB. In Lesson 10, the class models to explain how the nurse log helps animals to grow. In both lessons, students first model individually, then share ideas as a class. Given that the scale of these models shifts to include particles and energy (Lesson 7) and many new components (Lesson 10), it is probably easiest to model on a new piece of chart paper or digital space, with the previous versions of the model available as a reference for students to see how their ideas have changed.

Lesson 11: Termite Matter & Energy



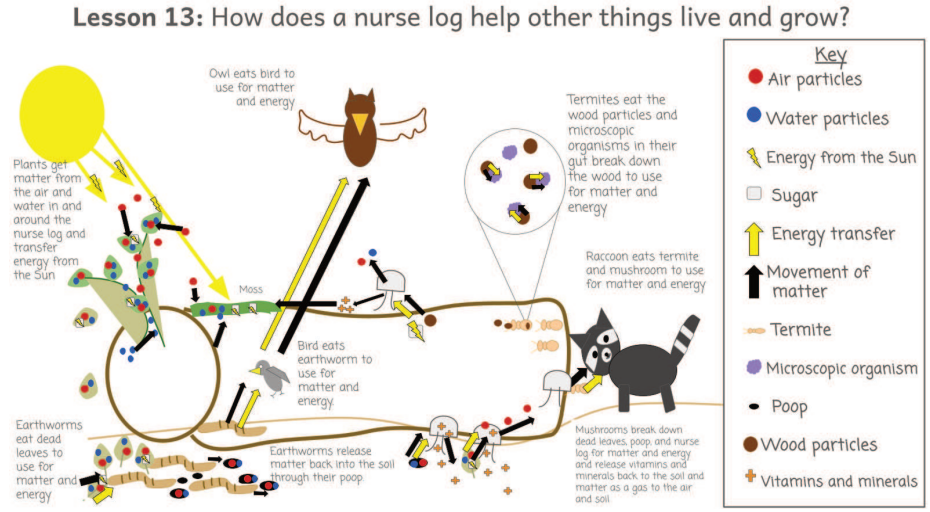
In Lesson 11, the class collaborates to model how termites get matter and energy from wood in the nurse log. Students share evidence from a variety of resources and identify how to use zoom-in circles to represent the different spatial scales of the process.

Lesson 13: Mushroom Matter & Energy



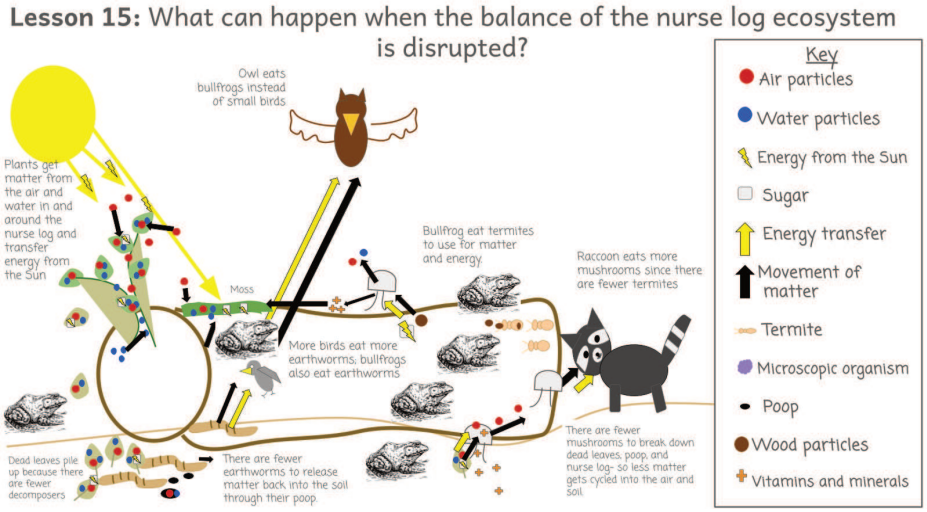
In the first part of Lesson 13, students work in pairs to model how fungi like mushrooms obtain and use matter and energy. This builds on what students have figured out about plants and animals, while identifying how fungi are different from these organisms.

Lesson 13: Final Consensus Model



Following the partner modeling to explain how mushrooms obtain and use matter and energy, students share their model ideas about mushrooms and fungi, as well as what they figured out in Lesson 11 about microscopic organisms inside of termites, as they revise the nurse log consensus model. At this point, the matter and energy pathways should be complete as a food web to show how matter and energy flow through the full nurse log system through feeding relationships. As in previous lessons, it may be helpful to create a new model and have the other versions of the model available as a reference.

Lesson 15: Disruptions to the Nurse Log Ecosystem



Before undertaking the final unit transfer task, the class adds bullfrogs to the final Class Consensus Model that they completed in Lesson 13. Students determine where bullfrogs fit into the food web and identify how bullfrogs disrupt the flows of matter and energy and impact the other organisms in the system. This is an opportunity to revisit and reinforce the ideas about bullfrogs and newly-introduced species that students figured out through a game in Lesson 14 to show the impacts of bullfrogs, which have been inadvertently introduced to nurse log ecosystems.

Discussion Type Prompts

Initial Ideas Discussion	Building Understandings Discussion	Consensus Discussion
When eliciting initial ideas: • What are your ideas to explain/solve ___? • Why/how do you think ___ happens? • How might we solve ___ problem? • Have you ever experienced something like this? How was that like this? • What did you notice about ___? • What questions do you have? **When clarifying and encouraging student interaction and pressing for reasoning:** • Can you share what you mean by "___"/(gesture)? • Can you tell us more about why you think that? • Who has a similar/different way of thinking about this topic? **When asking for or providing a synthesis of initial ideas:** • Who can summarize some of the initial ideas we've heard today? • So I'm hearing ___ and ___. Does that capture what we are currently thinking? Is there something I missed? • What new questions do we have? **When asking students how to investigate their initial ideas:** • What are some ways we could test our thinking? • What do we need to know to figure this out? • What could we do in our class to find answers to our questions?	**Setting and maintaining focus:** • Could someone tell us our question/problem we are trying to answer/solve? • We have a new piece of evidence. What does that mean for us? **When inviting a group to share:** • What are some of your ideas? • What are some important parts of your model? • Does this design do what we need it to do? How do we know? **When pushing for elaboration of evidence and reasoning:** • What's your evidence? • What evidence backs up what you are saying? • How does that idea fit with ___'s claim? • Can you tell us more about ___ part of your model/solution? Why did you draw/say/do that? Are you saying ___(restate the idea)? **When encouraging critique and student-to-student interaction:** • What questions do you have for your friends? • Does your evidence support your friends' ideas? If not, why? • Is there anything you would add? **To find tentative conclusions & next steps:** • What do we seem to agree upon? • What new questions do we have? • What else do we need to find out? What might we do next?	**During stock-taking:** • Could someone tell us the question/problem we are trying to answer/solve? • What are we thinking about our phenomenon? • What evidence supports our ideas? **When soliciting ideas to develop or modify the model or explanation:** • How should we represent it? Are we OK with that? • Do we agree? Do we need more evidence? • How are these explanations similar? How are they different? • How might we change our model or explanation to include those ideas? • What changes would you make to what you said/gestured? **When inviting support or critique:** • Are we missing anyone's ideas? • I'm hearing two different ideas: (restate ideas). What do we think about those? • Are there still places where we disagree? Can we clarify these? • Does it always work that way? **When soliciting ideas for next questions or investigations to pursue:** • What should we do next to figure out more about our questions? • What new questions do we have that might help us move forward?

Video Transcript: Redwood

Audio	Visual
0:00	White OpenSciEd Logo on a black background in the top right corner with the title “Fallen Redwood Tree” on the bottom and an orange banner that says “Student Video”
0:07	The camera moves from left to right across a large fallen redwood tree. There is a large wooden trunk that leads to leaves and branches as the camera moves to the left. The tree is in a forest and is surrounded by smaller trees.
00:15	The OpenSciEd Logo appears in white across a black screen